Brain-Inspired Hierarchical controller for Context-Dependent Motor Decision-Making in a Simulated Robotic Arm

Abstract

Biological motor control notably emerges from the interaction of multiple specialized neural structures that collectively integrate sensory information, but also evaluate behavioural context, action selection and generate movements. Computational models inspired by this organization thus can provide an opportunity to investigate how biological principles can be translated into adaptive robotic control. This project is largely inspired by the intersection between models of hierarchical control & computational neuroscience, presenting a modular brain-inspired controller for a two-degree-of-freedom robotic arm implemented in a MuJoCo physics engine. The framework will be evaluated through a series of experiments examining context-dependent motor decision-making and behavioural adaptation under changing task conditions. Results demonstrated that this proposed architecture produces coherent context-sensitive behaviour, or more broadly, illustrated how principles from computational neuroscience can be incorporated into interpretable robotic control systems

Introduction

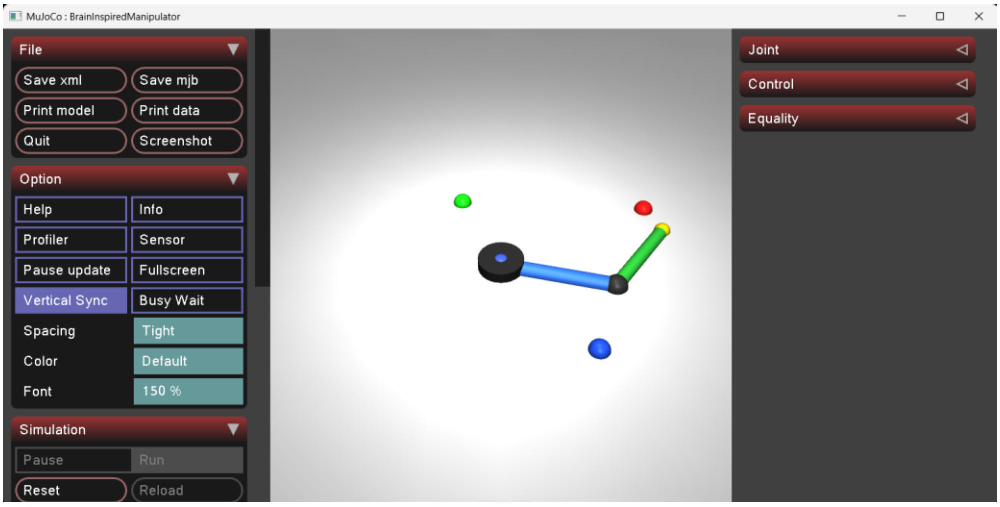
Controlling a physical body essentially requires solving tightly coupled problems of perception, decision-making, prediction and fast feedback while inevitably taking into account additional noise, delays and changing goals, problems that have prior been argued to be distributed across specialized brain-systems – cerebellum for fast supervised prediction, basal-ganglia for reinforcement-driven action-selection, and lastly the neocortex responsible for richer state representations and slow consolidation (e.g., Doya 1999; Kawato et al.). But translating these ideas into robotic controllers is still challenging since one has to uphold the neural hierarchy, not neglect biological plausibility and attempt to evaluate robustness in both static and dynamic environments. This project is attempting to address this gap via the implementation of a modular hierarchy that assembles cerebellum-style feedforward learners, basal-ganglia-like utility evaluation and stochastic selection, as well as cortical-like state representations, all of which converge into an embodied Mujoco arm controller. The specific contributions are hereby as follows: (1) A clear software abstraction in the form of BrainState that enforces hierarchical communication, (2) reproducible experimental infrastructure that enable systematic sweeps, (3) module representations that map into learning paradigms previously emphasized by Doya and lecture 10 of the Neuromorphic Control course, including a learnable cerebellar network and predictive trajectory leads for moving targets, and finally most importantly (4) three different experiments (top-down context modulation, graded virtual lesions, dynamic target tracking) coupled with subsequent analysis pipelines discussed later below. All of these steps together will aim to make the computational hypotheses from computational neuroscience testable in a controlled embodied setting.

Methodology

Overview:

The objective is to implement a biologically inspired modular sensorimotor hierarchy for a 2-degrees of freedom arm in Mujoco and subsequently run three different experiments (context modulation, graded module lesions, dynamic target robustness) to prove its functionality under varying circumstances. Top-Level modules hereby encompass: Vision, Parietal Cortex, Basal Ganglia, Motor Cortex, a trajectory generator, inverse kinematics, Spinal Cord and the Cerebellum. The codebase was organized in a way that each brain region is represented by one python module and all of them communicate via one BrainState module (brainstate.py). The numerical code uses, aside from regular python tools, Numpy/Pandas for data handling, as well as Pytorch for the cerebellar feedforward learner. The experiments are then configured via ExperimentConfig YAMLs and executed by generate_dataset_runner.py. The reproducibility is reinforced by deterministic per-module RGN's (utils) and experiment metadata that is being saved in the process (logger.py).

The Mujoco arm model is contained in the armscene.xml itself, which defines the planar 2-DoF manipulator (L1 = 0.40, L2 = 0.35), three target bodies with slide joints and site markers, as well as two motor actuators that are mapped to shoulder and elbow joints. The environment itself (environment.py) implements the target dynamics (static, sinusoidal, circular and random-walks, that are maintaining initial offsets so dynamic motion is relative to reset pose. They also provide external forces whenever enables manually and can compute difference-target velocities. Execution of the robot can be done via main.py, yet the particular conditions of the experimental type are largely contingent on the YAML files' variable instantiation.



Major design principles and mapping to neuroscience have been implemented so that computations motivated by Doya (1999) and the lecture 10 material are constructed as follows: cerebellum -> fast supervised feedforward predictor -> basal ganglia- > value estimation + reinforcement selection (utility computation), cortex/PFC -> contextual modulation and slow consolidation (executive gains/attention). Such a mapping has guided choices for learning rules and lesion models: there is fast, high-capacity but quickly updatable cerebellar learner, but also slower TD-style updates in the basal ganglia, and lastly the PFC as a top-down gain generator.

Other key Implementations

Lesions that are applied per module and are simulating natural degradations (attenuation + added noise) that the runner passes to each module if commanded. They are not binary removals computationally speaking here. Cerebellum learning is implemented as a small feedforward MLP that maps the local joint state and desired joints to a two-dimensional feedforward torque correction (delta t). Train_network.py computes and saves input/output normalizations, stores the weights in the models. Spinal PD controller handle low-level joint controls, with torques being computed as $\tau = Kp \odot (q\_des - q) + Kd \odot (dq\_des - dq)$, and Kd is chosen approximately as 2*sqrt(2) for critical damping. Action selection is inherent to the basal ganglia and is driven by a utility computation for each MotorPlan, which essentially combines PFC-modulated value and perceptual terms minus costs. There is likewise trajectory generation involved in using minimum-jerk time scaling to produce smooth position/velocity profiles, and lastly inverse kinematics that selected the solution nearest to the current posture (2-link solution: $c2 = (x^2+y^2-L1^2-L2^2)/(2L1L2)$, $q2 = \mathrm{atan2}(\pm\sqrt{(1-c2^2)}, c2)$, $q1 = \mathrm{atan2}(y,x) - \mathrm{atan2}(L2 \sin q2, L1+L2 \cos q2)$.

Specific Methods of the Experiments

Experiment 1

This one covers a contextual modulation through top-down PFC control, testing how behavioural context (REWARD-SEEKING; ENERGY-EFFICIENCY; EXPLORATION) change both target selection and movement. The PFC here maps context to executive signals (goal_bias, learning_gain, exploration_gain etc.) that downstream modules later read each timestep. For each context it runs multiple independent seeds each with episodes and steps configured in ExperimentConfig. The measured quantities hereby include chosen targets, selection probabilities, decision entropy, movement vigor, dopamine, learned values and energy (integral of $|\tau \cdot \omega|$). For the analysis the analysis_conditions.py is employed that aggregates episodes and runs inferential tests.

Experiment 2

The purpose of the graded lesion study is to quantify how graded/more or less biologically plausible degradations in individual modules (PFC, Parietal, BG, Cerebellum) even study eventually affect decision-making and motor performance. Lesion_sweep.py automates & differentiates between the degree of severity of the lesions (mild/moderate/severe). For each targeted module a LesionProfile is produced and a profile map is passed onto the dataset generator. The analysis plan includes aggregate episode summaries with lesion_analysis.py and compares the groups.

Experiment 3

The dynamic-target robustness & difficulty experiment assesses how predictive components (trajectory lead, cerebellar feedforward) will support the tracking of moving targets of varied difficulty, all while manipulating different motion types (static, sinusoidal, circular and randomized) and tracking target speeds & radii. The environment hereby provides the true target velocities, while vision computes finite-difference estimates which feed the TrajectoryGenerator's predictive lead. Optionally one can enable here random external forces or perturbations. Dynamic_analysis.py then computes episode summaries as in the other experiments it is done with their respective files, comparing dynamic vs static baselines and producing heatmaps of performance vs speed/radius).

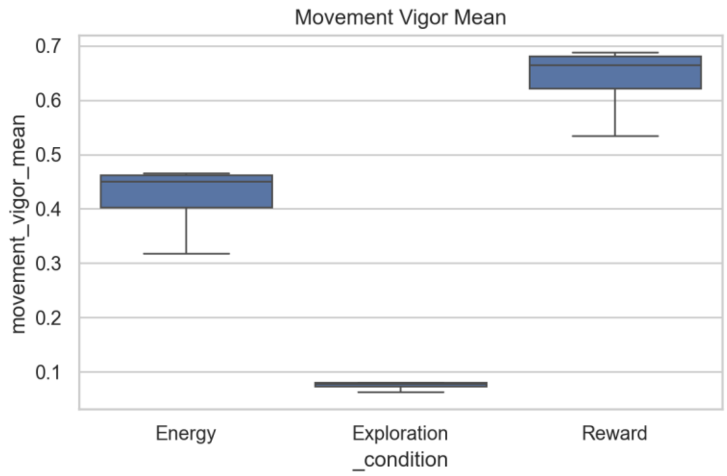
This methodology essentially stiches established building blocks (Mujoco simulation, PD control, closed-form IK, TD updates etc.) into a controlled embodies testbed that attempts to mirror the computational division of labor proposed in Doya (1999) and lecture 10, at the same time keeping the mapping from neuroscientific hypothesis -> code -> measurable outcome overall transparent for analysis & interpretation.

Results

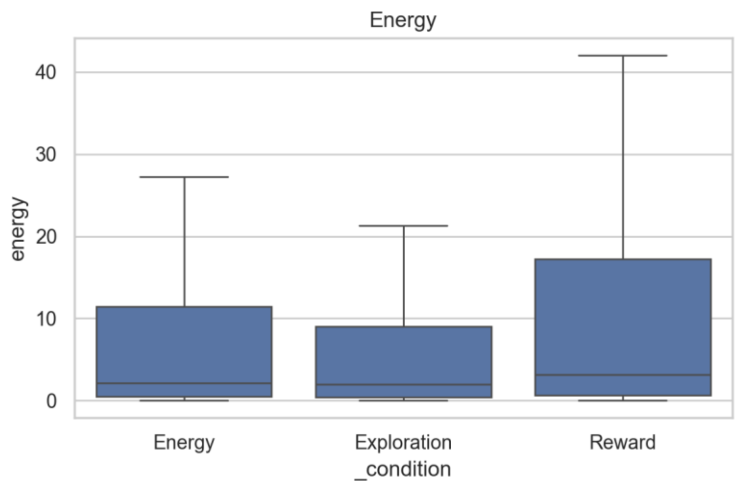
Experiment 1

The behavioural-context experiment was run which compared three PFC-controlled modes: Energy-Efficiency, Exploration, and Reward-Seeking, each run having independent samples (400 step/episode-level observations per condition) with a fixed experiment condition. Each condition produced hundreds of observations as well as non-parametric tests (Kruskal).

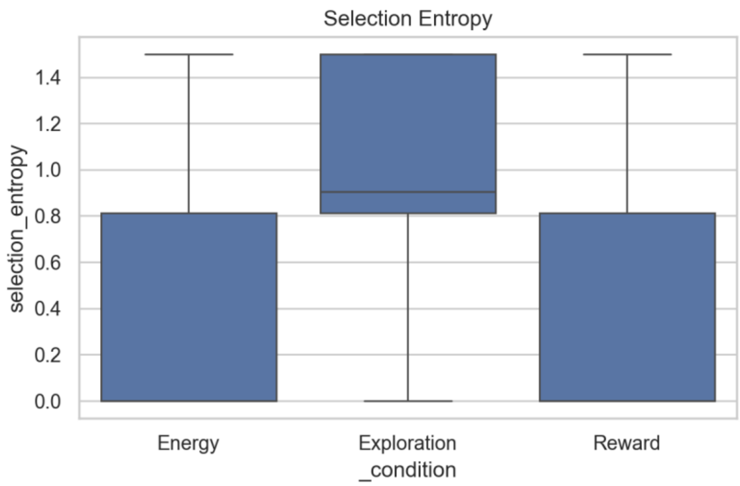
Main Findings



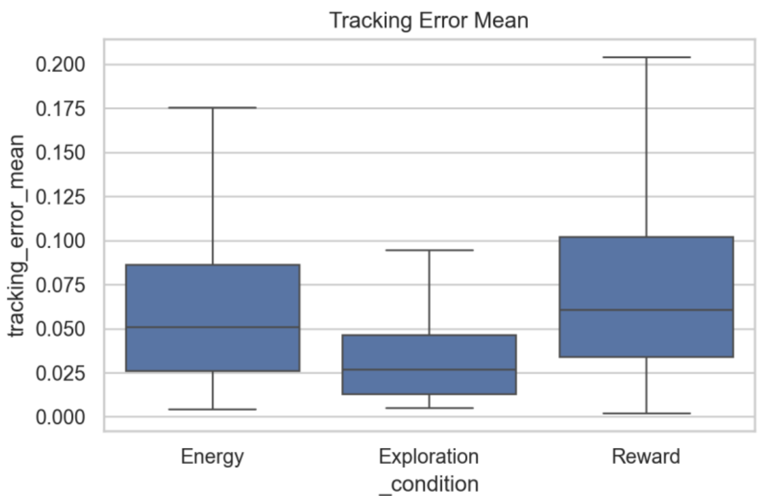
Reward-Seeking conditions produced substantially higher movement vigor than other contexts, with the boxplot displaying a median around ~0.6-0.7. This would be consistent with the PFC reward-biased executive signals increasing motor_gain and decision confidence which in turn raise movement vigor.



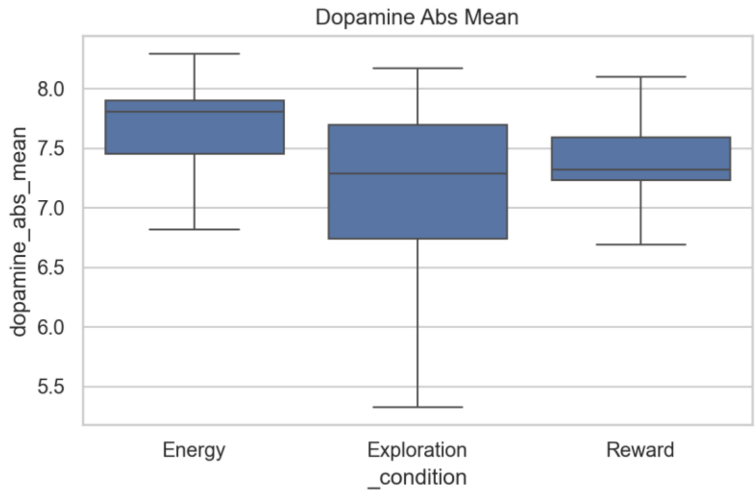
Energy consumption follows a similar yet much more even pattern, where the reward condition displays the largest spread and relatively higher median, whereas the energy condition is intermediate and exploration lower energy use. The energy metric differing across contexts was confirmed by non-parametric testing (Kruskal; p= 1.38e-4). A possible interpretation of this behaviour is that reward bias might trade energetic cost for speed/vigor. The energy condition reduces vigor and energy use as designed (PFC motor_gain lower), while exploration biases decisions differently.



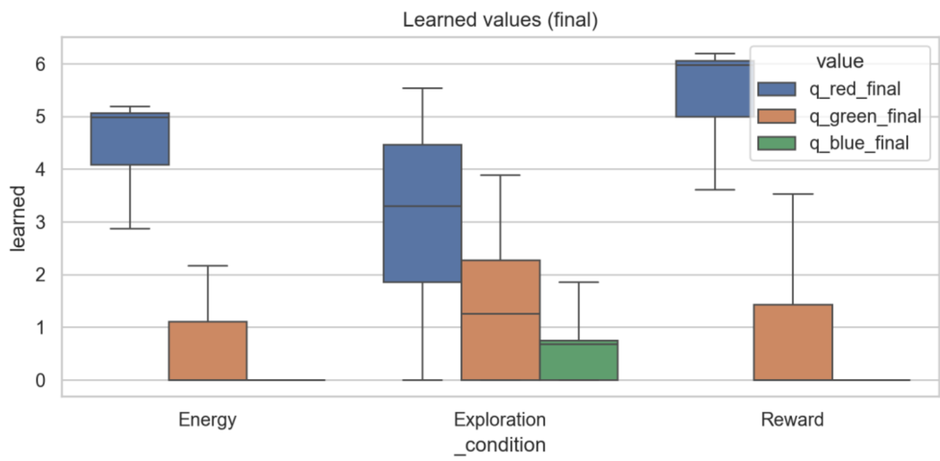
With regards to decision uncertainty/policy exploration, the selection entropy results in the highest value outcome under the exploration context and lowest within the energy & reward contexts. This pattern is likewise significant in the inferential table (produced by the analysis file as inferential_summary, p ≈ 3.3e-71). In other words, PFC modulation of exploration_gain increased policy stochasticity as was expected, given that that the system explored more under that context.



The sensorimotor accuracy is approximated by the tracking error (the rough measure of movement accuracy), which here is lowest in the exploration context, intermediate in energy, and highest in Reward. All related values are likewise statistically robust (Kruskal p ≈ 8.5e-37). The interpretation here is that higher vigor (reward context) produces faster but less accurate movements (speed-accuracy trade-off). Energy context seems to sacrifice some accuracy relative to Exploration in exchange for lower energy use.



The magnitude of prediction errors which drive learning updates also differ strongly between contexts (dopamine_abs_mean p ≈ 2.03e-51). This accords largely with differing experienced reward structure and policy behaviour, for example exploration producing more varied outcomes leading to larger or more variable prediction errors.



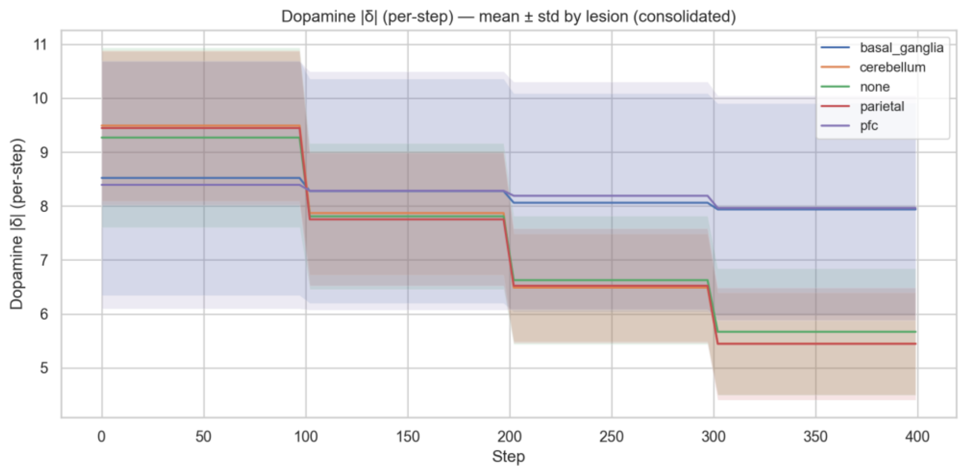
Across episodes the learned-values trajectory (target choice) differ by top-down context. Within the Reward context, the learned value for the high-reward target (here, red) naturally increases relative to others, as the system concentrates expected value on the better target). However in the Exploration context the learned values remain most similar and less discriminative which is likely consistent with higher selection entropy.

The overall verdict for experiment 1 is that manipulating PFC context produces the largely expected behavioural trade-offs: Reward context results in higher vigor and energy, selection confidence and yet worse tracking accuracy. Exploration context leads to highest selection entropy but lower vigor and lower tracking error (slower, more precise). Lastly, the Energy context results in lower energy and overall intermediate performance.

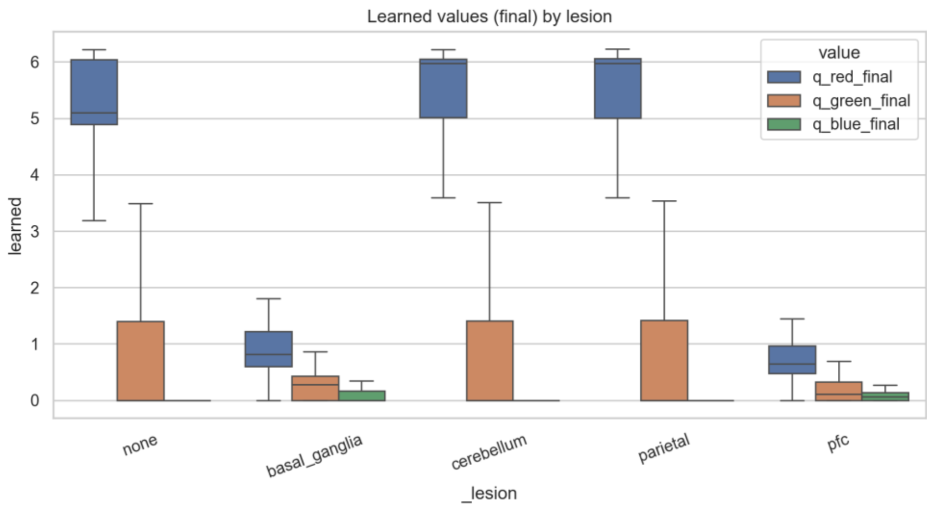
Experiment 2

The objective in the next experiment was to quantify how selective disruptions to different modules (PFC, parietal cortex, basal ganglia, cerebellum) affect decision-making, learning and motor performance in the same unchanged reaching task. The environment here is still held constant (static targets) and introduced controlled lesion profiles of varying severity to individual modules so the effect of each 'lesion' can be compared to the intact (control) network. Again, a sweep was run that produced many independent runs with random seeds for multiple lesion types as well as severity levels (mild/moderate/severe).

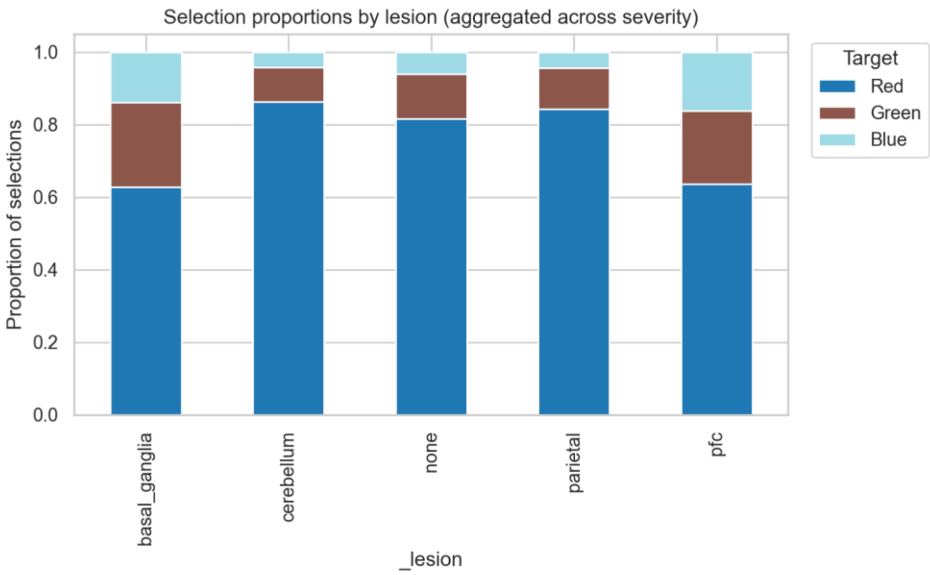
The decision confidence shows a summary of how certain agent's policy was when selecting an action, the value of which was computed as decision_confidence = 1-entropy/log2(number_of_options). The intact/control condition centers around 0.5, while PFC and BG-lesioned conditions formed a distinct low-confidence band, and parietal and cerebellum lesions sitting noticeably higher than the controls.



Dopamine PEs are scalar reward prediction errors used to drive learning and are located in the basal ganglia module as delta = reward – prediction, with the absolute value quantifying the surprise/magnitude of learning signals per step. The PFC & BG- lesioned runs exhibit the largest |δ| especially later on as the episodes progress. The clear stepwise drops near the episode boundaries could reflect learning/stabilization across episodes.



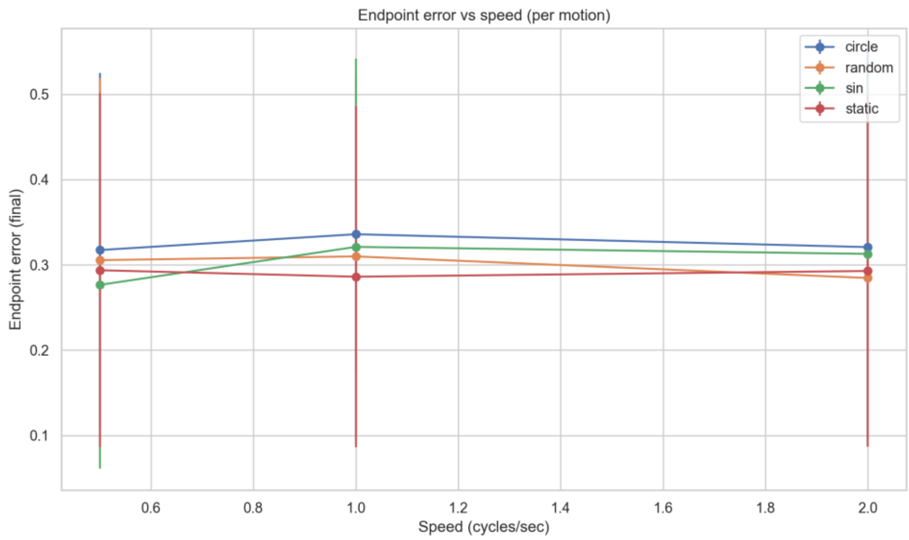
These are the final per-episode learned value estimates stored for each target. An empirical pattern here is that the intact group shows high median learned value for the high-reward target (RED), which is reflected in the Cerebellum and parietal lesions as well. PFC and BG lesions display substantially lower learned-value magnitudes.

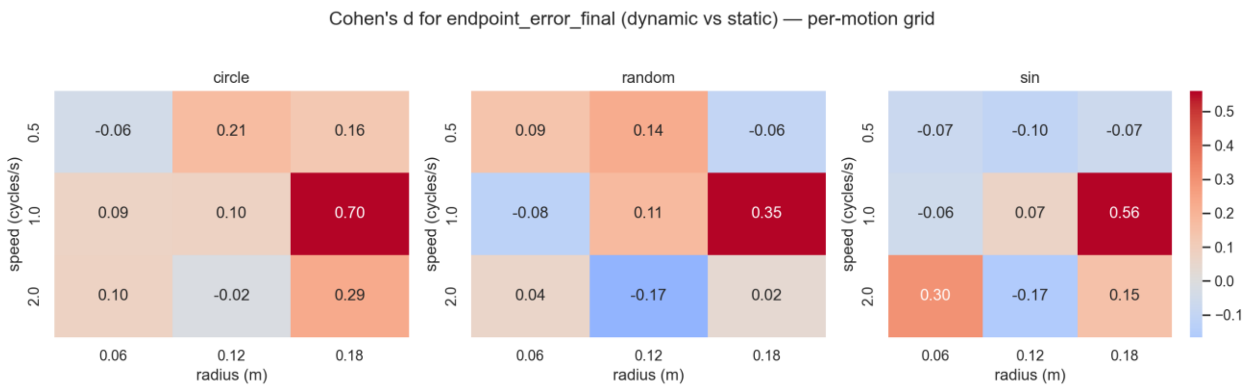


Lastly the normalized proportions of target choices (Red, Green, Blue) aggregated across runs, grouped by lesion type. Again, most groups naturally still primarily select Red (highest reward target), yet the BG and PFC lesion conditions show higher fractions of non-Red selections compared with control and cerebellum/parietal lesion groups.

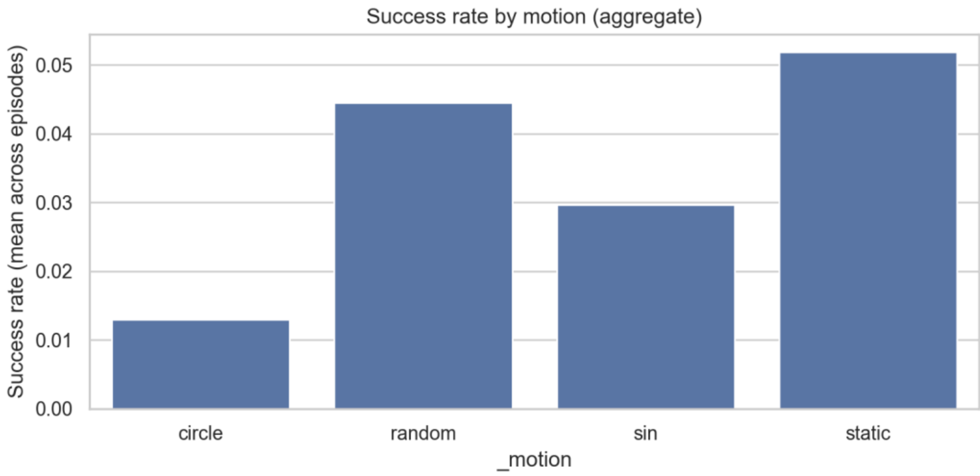
Experiment 3

The objective here was to test whether the architecture's predictive components (finite-difference target-velocity from Vision, the TrajectoryGenerator's predictive lead, and the cerebellar feedforward correction) all improve tracking when targets move with different motion types, those being sinusoidal, circular and random movements, across a grid of speeds & target radii, compared always to the static baseline as utilized in the former two experiment procedures. The dynamic target behaviour and its respective computations depending on motion type is located in environment.py.





Within the evaluation of the aggregate endpoint-error vs speed per motion trends one can observe the endpoint-error rising only modestly for most dynamic conditions relative to static. Circle motion usually shows the highest final endpoint error across speeds, random and sinusoidal are closer to default (static) on average. The overall differences across speeds are small in the group means, which would suggest that the controller maintains roughly similar accuracy over the tested speed range for many conditions. This is likewise reflected in Cohen's d grid which largely mirrors the delta heatmaps but in standardized units. Interestingly specific motion/scale combinations reliably degrade performance but improve again once a further speed increase is applied (from 1.0 to 2.0).



The success rate describes the fraction of episodes ending with the endpoint threshold, here with static trials showing the highest mean success (~5% across episodes), followed by random, then sinusoidal and lastly circle. One can conclude that reaching within the endpoint threshold is already rare overall, since thresholds are stringent and episodes are short, yet the rates here follow the somewhat expected ordering by difficulty and corroborate the endpoint-error results, as motion will increase the failure likelihood, particularly for circular motion at challenging speed/radii.

Discussion

Were the expectations met?

The project largely achieved its first two aims, that of a modular BrainState-based codebase with per-module RNGs and metadata (logger) that makes runs reproducible and easy to sweep, all while producing large datasets and analysis/evaluation results. The three-module mapping inspired by Doya/Lecture 10 (PFC-> top-down gains; BG -> value + stochastic selection; Cerebellum -> fast feedforward prediction) is in large supported by the experimental patterns that were observable. For example, maipulating PFC parameters produced the expected trade-offs in vigor, energy & selection entropy, the targeted lesions in BG & PFC changed selection statistics and learning signals in ways that should be consistent with their roles, and adding movement dynamics highlighted again the benefit and limits of short-horizon predictive components.

Experiment 1

The PFC-controlled contexts produced the predicted behavioural trade-offs, in that reward context largely increased movement Vigor and energy, Exploration increased the entropy & reduced vigor/tracking error, while Energy contexts reduced vigor & energy by trading off some accuracy. This result may very well align with the theoretical role of the prefrontal cortex as a top-down gain controller (Lecture 10 context -> executive gains), and it validates that the high-level control knobs such as goal-bias, learning_gain, exploration_gain and motor_gain propagate to downstream modules and other behaviour.

From a mechanistic interpretation with regards to the selection/decision layer, increasing the PFC's exploration_gain appears to raise the effective temperature in the SoftMax selection (Basal Ganglia), which produces the higher selection entropy observable. It's the textbook prediction from the hierarchical control hypothesis, in that PFC sets an exploration regime by up-weighting temperature, in turn increasing the stochastic sampling of alternatives. Likewise, the higher motor_gain under the Reward regime leading to larger movement vigor and inferred energy (integral of $|\tau \cdot \omega|$) is consistent with PFC boosting motor gain to favor speed and immediate reward capture.

The interaction between the Exploration condition and its speed vs accuracy tradeoff also shows higher endpoint error, interestingly, suggesting that in this hierarchical architecture lowering motor_gain alone does not guarantee better endpoint accuracy if other factors such as choice variability, learning stability and perceptual certainty are changing concurrently. After all, high selection entropy means that agents frequently switch targets or policies across episodes, so this might reduce the amount of consistent reinforcement experienced for any one target, producing shallower learned-value differences and therefore poorer global stabilization.

Some caveats to bear in mind include the present experiments using a specific mapping from context to gain values that were chosen to produce visible behaviour; so different quantitative mappings could change the qualitative outcomes. The reward schedule in this environment is also non-stationary (phased changes over time), so it might interact with Exploration to shape learning dynamics.

Experiment 2

A plausible interpretation for the results observed in the lesion experiment is certainly the PFC acting as a top-down gain and certainty generator. Its lesion model reduces gain_scale and learning/attention parameters, thus the executive gains are attenuated, while downstream systems loose motor drive (low movement_vigor) and confidence. On the other hand, the basal ganglia module's SoftMax operates at higher effective temperature and choices become more random (higher entropy), a randomness that creates more variable outcomes and larger learning signals (|δ|). BG lesions primarily degraded the value-comparison and SoftMax selection computations, where utilities are flattened and noisy, so choices diversify and learning signals grow. Lastly, cerebellar lesions mainly impair fast predictive compensation and feedforward correction, the reduction of which leads to the spinal cord/PD controller and motor cortex to compensate with larger torques or less efficient commands – hence higher entropy and energy trackings. The still dominant Red bias under cerebellum lesions suggests reward/value learning in BG still operating so selection remains very reward-directed, but the motor execution simply becomes less efficient.

There are nevertheless visible points of tension, in that some metrics pull in different directions, in that cerebellum-lesioned runs sometimes had relatively modest endpoint errors while showing much larger tracking errors and energy use. This can point towards endpoint_final alone being able to mask unstable or energetically costly control, and overall that different error definitions definitely point here to different conclusions in the temporal analyses. Also, PFC lesions produced an extremely large drop in movement vigor (mean around 0.04), which might be plausible given the PFC module's role in motor gain in the mapping, but it also amplifies downstream interpretation difficulties. What was learned here is how multidimensional behavioural metrics can be; designing such experiments that separate motor execution from decision-making components is hard but crucial.

Experiment 3

Lastly, the dynamic-target sweep shows a nuanced picture, in that predictive components such as finite-difference vision and trajectory-generator leads help in many conditions, yet benefits depend here on the motion type, speed & workspace radius. It becomes clear that predictability matters, as for smooth, periodic motions (sinusoidal/circle) a modest predictive lead provides a useful head-start (lead_time ~ 0.25×nominal duration was used) that reduces tracking error and sometimes endpoint error. Random motion is naturally also less predictable, inherently, since the finite-difference velocity estimates from the vision module are instantiated as noisy and lagged. The gains are thus smaller and unreliable.

Furthermore, a peculiar, non-monotonic speed effect was observed f.ex via above heatmaps, where the endpoint error often peaks at a speed of 1.0 and drops again at speed = 2.0. Such counterintuitive non-monotonicity could suggest an interaction between (a) the target's temporal frequency or (b) the movement planning timescale (trajectory duration). A possible velocity estimation/noise hypothesis could be established given that Vision computes target velocity by finite differences. At moderate speeds than (1.0) these finite-difference estimates may be both large enough to require meaningful lead and noisy enough (or biased by delay/occlusion) that the lead amplifies the error. At higher speeds however (2.9) the controller may shorten nominal durations (clamped by min/max limits) or rely less on the same relative lead fraction.

Overall, it can be concluded that experiment 3 showed the controller already handling many dynamic conditions with only modest degradations and the system seems to preserve largely the learned-value structure, so decision-level learning is quite robust. But predictable pockets of failure highlight that the current predictive lead and cerebellar feedforward are only partially effective for moderately fast and large-radius motion (where target displacement during the nominal lead is large). To close these gaps it would likely be best to address perception (aka velocity estimation), the adaptive prediction and especially cerebellar training.

References

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Doya, K. (1999). What are the computations of the cerebellum, the basal ganglia and the cerebral cortex? *Neural Networks, 12*(7-8), 961–974. https://doi.org/10.1016/S0893-6080(99)00046-5

Disclaimer on AI usage: Github copilot was utilized to make sure of the interconnectedness between the multitude of modules as well as internal consistency of the py files themselves.